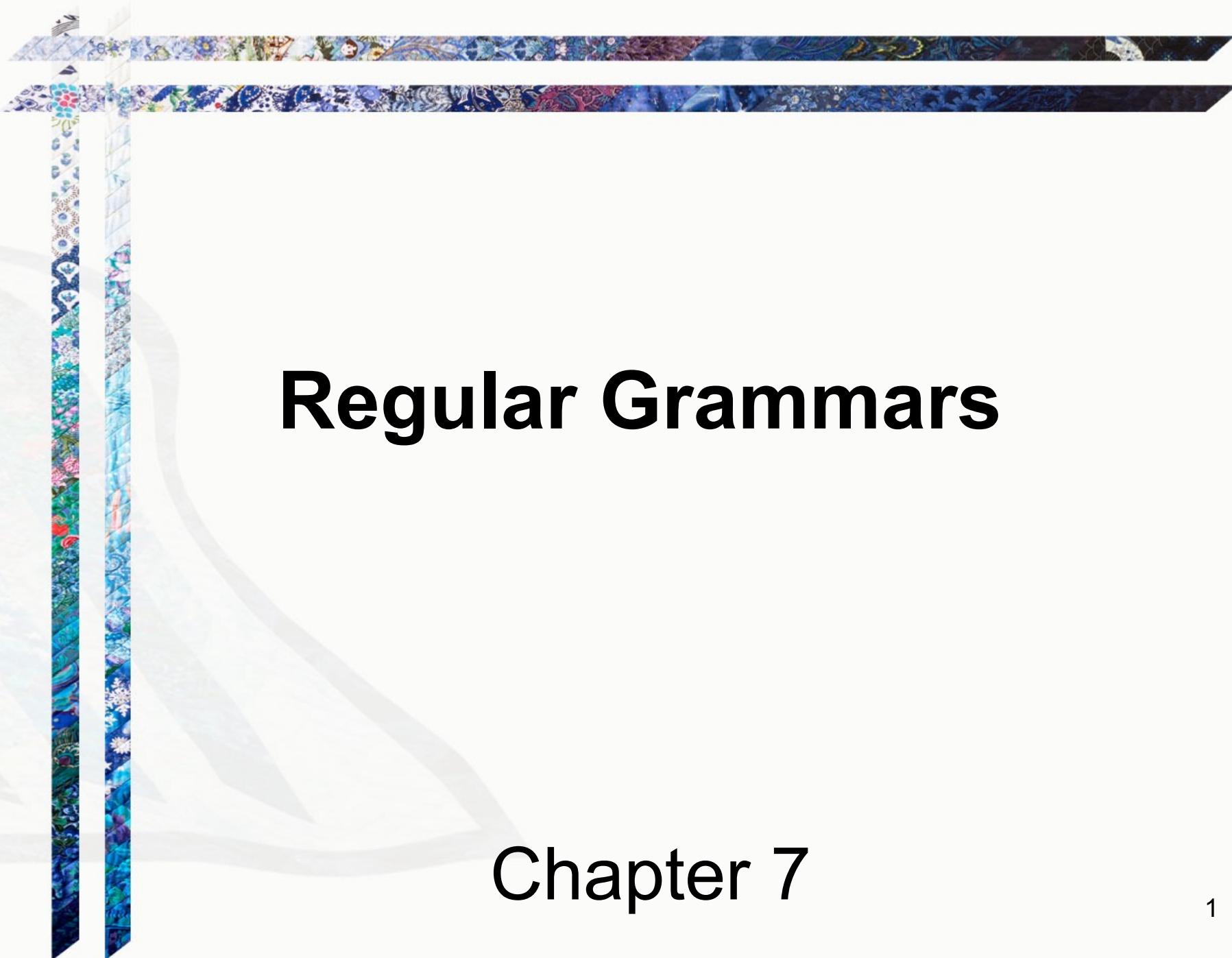




Regular Grammars

Chapter 7

Grammars

A **grammar** G is a quadruple (V, Σ, R, S) , where:

- V is the rule alphabet, which contains **nonterminals** and **terminals**,
- Σ (the set of terminals) is a subset of V ,
- R is a finite set of **rules** of the form:

$$X \rightarrow Y, \quad X, Y \in V^*$$

- $S \in V - \Sigma$ -- the **start symbol**

Regular Grammars

In a **regular grammar**, all **rules** in R must:

- have a **left hand side** that is a single nonterminal
- have a **right hand side** that is:
 - ε , or
 - a single terminal, or
 - a single terminal followed by a single nonterminal.

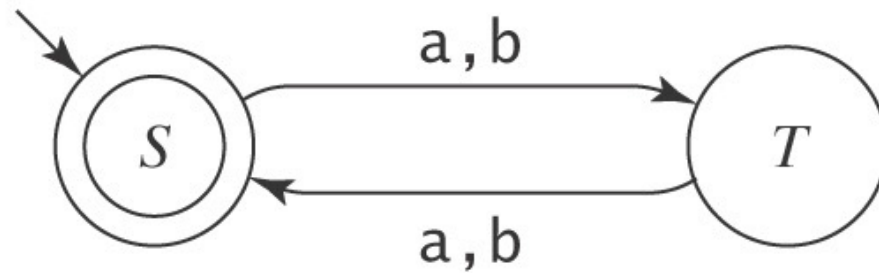
Legal: $S \rightarrow a$, $S \rightarrow \varepsilon$, and $T \rightarrow aS$

Not legal: $S \rightarrow aSa$ and $aSa \rightarrow T$

The **language** defined by a grammar: all terminal strings that can be obtained starting from S and applying the rules

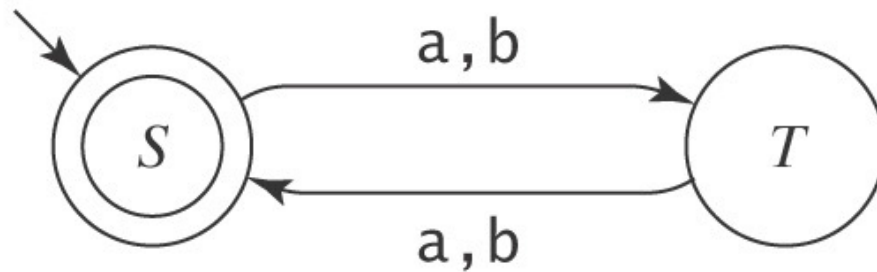
Regular Grammar Example

$$L = \{w \in \{a, b\}^* : |w| \text{ is even}\} \quad ((aa) \cup (ab) \cup (ba) \cup (bb))^*$$



Regular Grammar Example

$L = \{w \in \{a, b\}^* : |w| \text{ is even}\} \quad ((aa) \cup (ab) \cup (ba) \cup (bb))^*$



Grammar:

$S \rightarrow \varepsilon$
 $S \rightarrow aT$
 $S \rightarrow bT$
 $T \rightarrow a$
 $T \rightarrow b$
 $T \rightarrow aS$
 $T \rightarrow bS$

Language:

$S \Rightarrow bT \Rightarrow bb$
 $S \Rightarrow aT \Rightarrow abS \Rightarrow ab bT$
 $\quad \Rightarrow abbaS \Rightarrow abba$
 $S \Rightarrow \varepsilon$



Regular Languages and Regular Grammars

Theorem: The class of languages that can be defined with regular grammars is exactly the regular languages.

Proof: By two constructions.

Regular Languages and Regular Grammars

Regular grammar $\rightarrow$ FSM:

grammartofsm($G = (V, \Sigma, R, S)$) =

1. Create in M a separate state for each nonterminal in V .
2. Start state is the state corresponding to S .
3. If there are any rules in R of the form $X \rightarrow a$, for some $a \in \Sigma$, create a new state labeled $\#$.
4. For each rule of the form $X \rightarrow a Y$, add a transition from X to Y labeled a .
5. For each rule of the form $X \rightarrow a$, add a transition from X to $\#$ labeled a .
6. For each rule of the form $X \rightarrow \varepsilon$, mark state X as accepting.
7. Mark state $\#$ as accepting.

FSM $\rightarrow$ Regular grammar: Similarly.



Strings that End with *aaaa*

$L = \{w \in \{a, b\}^* : w \text{ ends with the pattern } aaaa\}.$

$S \rightarrow aS$

$S \rightarrow bS$

$S \rightarrow aB$

$B \rightarrow aC$

$C \rightarrow aD$

$D \rightarrow a$

Strings that End with *aaaa*

$L = \{w \in \{a, b\}^* : w \text{ ends with the pattern } aaaa\}.$

$S \rightarrow aS$

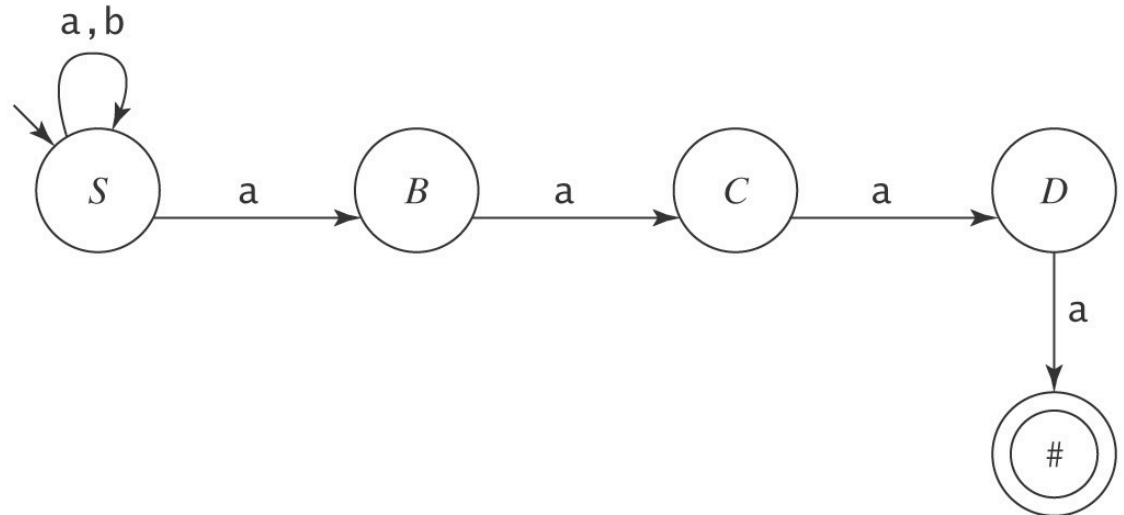
$S \rightarrow bS$

$S \rightarrow aB$

$B \rightarrow aC$

$C \rightarrow aD$

$D \rightarrow a$



Example 2 – One Character Missing

$S \rightarrow \varepsilon$

$S \rightarrow aB$

$S \rightarrow aC$

$S \rightarrow bA$

$S \rightarrow bC$

$S \rightarrow cA$

$S \rightarrow cB$

$A \rightarrow bA$

$A \rightarrow cA$

$A \rightarrow \varepsilon$

$B \rightarrow aB$

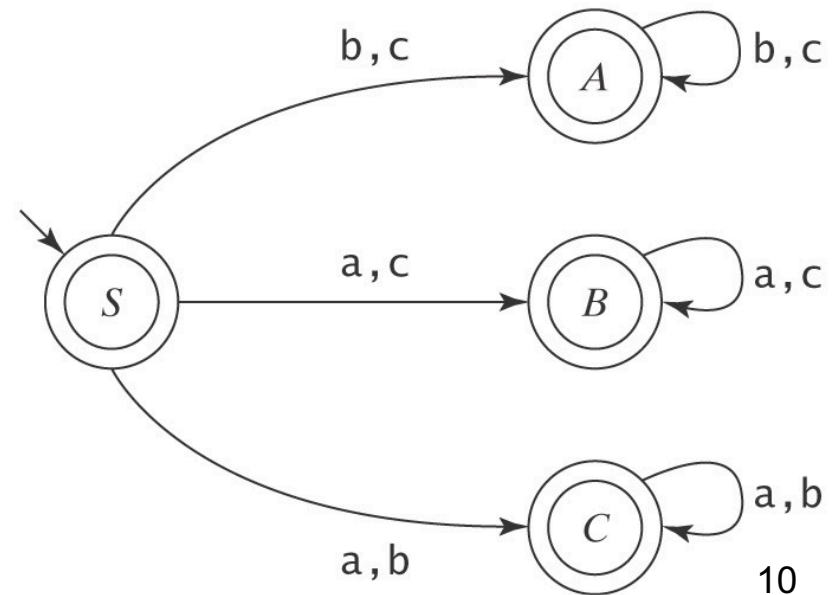
$B \rightarrow cB$

$B \rightarrow \varepsilon$

$C \rightarrow aC$

$C \rightarrow bC$

$C \rightarrow \varepsilon$



Conversions

